Operating Systems – 2022 Fall

Part II. – Questions

1) What are the typical attributes of a directory? (3 points)
   (Describe them with one sentence for full mark!)

2) What are the preemptive and non-preemptive strategies for process scheduling? Shortly describe them! (4 points)

3) What are the two fundamental models of inter-process communication? (4 points)
   (Describe them with several sentences for full mark!)

4) What is a system call? (1 point)

5) What is paging? (2 points)

6) What is the readers-writers problem? (2 points)

7) What is the CPU–I/O Burst Cycle? (2 points)

8) Just as a file must be opened before it is used, a file system must be ________ before it can be available to processes on the system (1 point)

9) What are the three requirements a solution to the critical-section problem must satisfy?
   (Describe them with one or two sentences for full mark!) (6 points)

10) What is the term for describing the situation where a waiting process is never again able to change state, because the resources it has requested are held by other waiting processes? (1 point)
    ____

Evaluation

Part I. - Test : at least seven (7) good answers required for grading the exam!

Part II. - Questions:

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are the typical attributes of a file? (4 points)
ribe them with one sentence for full mark!)

t are the strategies for disk scheduling?
oint)
ose one and shortly describe it! (2 points)

at are the two fundamental models of inter-process communication? (4 points)
escribe them with several sentences for full mark!)

hat is a system call? (1 point)

Vhat is segmentation? (2 points)

What is the problem if all philosophers simultaneously pick up their left fork? (2 points)

What kernel data structure can be used for one technique of passing parameters to system calls? (2 points)

Breaking logical memory into blocks of the same size called Internal fragmentation. (1 point)

What are the three requirements a solution to the critical-section problem must satisfy? (Describe them with one or two sentences for full mark!) (6 points)

What is the term for describing the situation where a process can be temporarily out of memory to a backing store and then brought back into memory for continued execution? (1 point)

critical section

7)

9)

luation

I. - Test : at least seven (7) good answers required for grading the exam!

II. - Questions:

12	failed
16	passed
19	average

Operating Systems – 2022 Fal

Part II. – Questions

1) What are the four components of a process? (4 points)
(Describe them with one or two sentences for full mark!)

2) What are the three strategies for selecting a free hole from the set of available holes? (1 point)
Choose one and shortly describe it! (2 points)

3) What are the two fundamental models of inter-process communication? (4 points)
(Describe them with several sentences for full mark!)

4) What is a system call? (1 point)

5) What is internal fragmentation and where could it happen? (2 points)

6) What is the problem if all philosophers simultaneously pick up their left fork? (2 points)

7) What kernel data structure can be used for one technique of passing parameters to system calls? (2 points)

8) A program that has been loaded and executing is called a __________. (1 point)

9) What are the three requirements a solution to the critical-section problem must satisfy?
(Describe them with one or two sentences for full mark!) (6 points)

10) What is the term for describing the situation where shared data may be manipulated concurrently and the outcome of the execution depends upon the order of access? (1 point)

Operating Systems – 2022 Fall

Part II. – Questions

1) What are the typical attributes of a directory? (3 points)
   (Describe them with one sentence for full mark!)

2) What are the preemptive and non-preemptive strategies for process scheduling? 3)
   Shortly describe them! (4 points)

3) What are the two fundamental models of inter-process communication? (4 points)
   (Describe them with several sentences for full mark!) 4)

4) What is a system call? (1 point)

5) What is paging? (2 points)

6) What is the readers-writers problem? (2 points) 9)

7) What is the CPU–I/O Burst Cycle? (2 points)

8) Just as a file must be opened before it is used, a file system must be ________ before it can be available to processes on the system (1 point)

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   (Describe them with one or two sentences for full mark!) (6 points)

10) What is the term for describing the situation where a waiting process is never again able to change state, because the resources it has requested are held by other waiting processes? (1 point)

    ____

Evaluation

Part I. - Test : <u>at least seven (7) good answers</u> required for grading the exam!

Part II. - Questions:

are

By understanding this declaration, I admit that I have noticed that any kinds of devices or materials are NOT allowed and NOT acceptable on this exam, and I agree to that. I accept all the ethical rules of the University.

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signature

Part I. – Test

1. A process can be terminated due to ________
   A) normal exit
   B) fatal error
   C) killed by another process
   D) all of the mentioned

2. A set of processes is deadlock if ________
   A) each process is blocked and will remain so forever
   B) each process is terminated
   C) all processes are trying to kill each other
   D) none of the mentioned

3. If a process fails, most operating system write the error information to a ________
   A) log file
   B) another running process
   C) new file
   D) none of the mentioned

4. In operating system, each process has its own ________
   A) address space and global variables
   B) open files
   C) pending alarms, signals and signal handlers
   D) all of the mentioned

...the next instruction to be executed by the current process is provided by the ________

A) Program counter
B) CPU registers
C) Pipe
D) Process stack

6. To access the services of operating system, the interface is provided by the ________
   A) System calls
   B) API
   C) Library
   D) Assembly instructions

7. What is inter-process communication?
   A) communication between two process
   B) communication within the process
   C) communication between two threads of same process
   D) none of the mentioned

8. What is operating system?
   A) collection of programs that manages hardware resources
   B) system service provider to the application programs
   C) link to interface the hardware and application programs
   D) all of the mentioned

9. What is the ready state of a process?
   A) when process is waiting to be scheduled to run after some execution
   B) when process is unable to run until some task has been completed
   C) when process is using the CPU
   D) none of the mentioned

10. Which one of the following is NOT true?
    A) kernel is made of various modules which cannot be loaded in running operating system
    B) kernel is the program that constitutes the central core of the operating system
    C) kernel is the first part of operating system to load into memory during booting
    D) kernel remains in the memory during the entire computer session

Operating Systems – 2021 Fall

Part II. – Questions

1) What are the advantages of multiprogramming? (4 points)
   (Describe just four of it with one or two sentences for full mark!)

2) What are the three strategies for selecting a free hole from the set of available holes? (1 point)
   Choose one and shortly describe it! (2 points)

3) What are the fundamental components of the Process Control Block? (4 points)
   (Describe a few of it with sentences for full mark!)

4) What is a context switch? (1 point)

5) What is external fragmentation and where could it happen? (2 points)

6) What is the problem if all philosophers simultaneously pick up their left fork? (2 points)

7) What is Shortest Remaining Time, SRT scheduling? (2 points)

8) A program that has been loaded and executing is called a ____________. (1 point)

9) Explain semaphores with details!
   (Describe its types, operations and their short pseudo-code for full mark!) (6 points)

10) What is the term for describing the situation where shared data may be manipulated concurrently and the outcome of the execution depends upon the order of access? (1 point)

uation

Test : at least six (6) good answers required for grading the exam!

Questions:

failed
passed
average
good
excellent

Operating Systems – 2021 Fall

Part II. – Questions

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Questions:

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passed
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excellent

22 Fall

08. December 2022.

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5. The address of the next instruction to be executed by the current process is provided by the ____

A) Program counter
B) CPU registers
C) Pipe
D) Process stack

6. To access the services of operating system, the interface is provided by the ____

A) System calls
B) API
C) Library
D) Assembly instructions

7. Why might an operating system include a disk defragmentation utility?

A) To change the size of partitions on the disk
B) To repair damaged sectors on a disk
C) To reorganize small amounts of unused disk space
D) none of the mentioned

8. How does the operating system manage programs that are running?

A) It allocates free memory when a program requests it.
B) It deletes programs from disk once they have been loaded into memory.
C) It converts programs to machine code and begins their execution.
D) all of the mentioned

9. What is the blocek state of a process?

A) when process is waiting to be scheduled to run after some execution
B) when process is unable to run until some task has been completed
C) when process is using the CPU
D) none of the mentioned

10. Which one of the following is NOT true?

A) kernel is made of various modules which cannot be loaded in running operating system
B) kernel is the program that constitutes the central core of the operating system
C) kernel is the first part of operating system to load into memory during booting
D) kernel remains in the memory during the entire computer session

Operating Systems – 2022 Fall

[illegible] December 2022

Name: [illegible] Neptun ID: [illegible]

Declaration

By understanding this declaration, I admit that I have noticed that any kinds of devices or materials are NOT allowed and NOT acceptable on this exam, and I agree to that. I accept all the ethical rules of the University.

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Part I. – Test

1. Which of these is not a job of the operating system?________
   A) Stopping unauthorized users accessing files that do not belong to them.
   B) Handling HTTP requests.
   C) Allocating memory to programs that are running on the computer.
   D) all of the mentioned

2. A set of processes is deadlock if________
   A) each process is blocked and will remain so forever
   B) each process is terminated
   C) all processes are trying to kill each other
   D) none of the mentioned

3. If a process fails, most operating system write the error information to a________
   A) log file
   B) another running process
   C) new file
   D) none of the mentioned

4. In operating system, each process has its own________
   A) address space and global variables
   B) open files
   C) pending alarms, signals and signal handlers
   D) all of the mentioned

22 Fall

08. December 2022.

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5. The address of the next instruction to be executed by the current process is provided by the ______
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    - D) kernel remains in the memory during the entire computer session

Operating Systems – 2021 Fall

Part II. – Questions

1) What are the advantages of multiprogramming? (4 points)
   (Describe just four of it with one or two sentences for full mark!)

2) What are the three strategies for selecting a free hole from the set of available holes? (1 point)
   Choose one and shortly describe it! (2 points)

3) What are the fundamental components of the Process Control Block? (4 points)
   (Describe a few of it with sentences for full mark!)

4) What is a context switch? (1 point)

5) What is external fragmentation and where could it happen? (2 points)

6) What is the problem if all philosophers simultaneously pick up their left fork? (2 points)

7) What is Shortest Remaining Time, SRT scheduling? (2 points)

8) A program that has been loaded and executing is called a ____________. (1 point)

9) Explain semaphores with details!
   (Describe its types, operations and their short pseudo-code for full mark!) (6 points)

10) What is the term for describing the situation where shared data may be manipulated concurrently and the outcome of the execution depends upon the order of access? (1 point)

uation

Test : at least six (6) good answers required for grading the exam!

Questions:

failed
passed
average
good
excellent

Operating Systems – 2021 Fall

Part II. – Questions

1) What are the advantages of multiprogramming? (4 points)
   (Describe just four of it with one or two sentences for full mark!)

2) What are the three strategies for selecting a free hole from the set of available holes? (1 point)
   Choose one and shortly describe it! (2 points)

3) What are the fundamental components of the Process Control Block? (4 points)
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5) What is external fragmentation and where could it happen? (2 points)

6) What is the problem if all philosophers simultaneously pick up their left fork? (2 points)

7) What is Shortest Remaining Time, SRT scheduling? (2 points)

8) A program that has been loaded and executing is called a ____________. (1 point)

9) Explain semaphores with details!
   (Describe its types, operations and their short pseudo-code for full mark!) (6 points)

10) What is the term for describing the situation where shared data may be manipulated concurrently and the outcome of the execution depends upon the order of access? (1 point)

uation

Test : <u>at least six (6) good answers</u> required for grading the exam!

Questions:

failed
passed
average
good
excellent

Ope[redacted]

Name: [redacted]

Declaration

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signature

Part I. – Test

1. A process can be terminated due to ________
   A) normal exit
   B) fatal error
   C) killed by another process
   D) all of the mentioned

2. If a deadlock is happened, the OS should always kill the process in the deadlock.
   A) true
   B) false

3. Which of the following scheduling algorithms use Time Quantum?
   C) FCFS
   D) SJF
   E) Round Robin
   F) Priority Scheduling

4. In operating system, each process has its own ________
   A) address space and global variables
   B) open files
   C) pending alarms, signals and signal handlers
   D) all of the mentioned

5. The state of a process is stored in its ________.
   A) PCB
   B) CPU registers
   C) Memory
   D) Process stack

6. Paging avoids internal memory fragmentation.
   A) True
   B) False

7. What is inter-process communication?
   A) communication within the process
   B) communication between two process
   C) communication between two threads of same process
   D) none of the mentioned

8. In multicore systems, asymmetric multiprocessing uses one core to handle the scheduling for all the other cores.
   A) true
   B) false

9. From blocked state, a process can only enter into ________.

   A) Running state
   B) Ready state
   C) New state
   D) Terminated state

10. Which one of the following is NOT true?
    A) kernel is made of various modules which cannot be loaded in running operating system
    B) kernel is the program that constitutes the central core of the operating system
    C) kernel is the first part of operating system to load into memory during booting
    D) kernel remains in the memory during the entire computer session

Operating Systems – 2022 Fall

Part II. – Questions

1) What are the typical instruction-execution cycle steps for the von Neumann architecture? (4 points)
   (Describe it with full sentences for full mark!)

2) What is the dual-mode operation of the Operating System?
   (Describe it with full sentences for full mark!) (3 points)

3) What are the fundamental components of the Process Control Block? (4 points)
   (Describe a few of it with sentences for full mark!)

4) What is a context switch? (1 point)

5) What is internal fragmentation and where could it happen? (2 points)

6) Describe the Critical-Section problem with your own example! (2 points)

7) What is Round-Robin scheduling? (2 points)

8) A program that has been loaded and executing is called __________. (1 point)

9) Explain semaphores with details!
   (Describe its types, operations and their short pseudo-code for full mark!) (6 points)

10) What is the term for describing the situation where shared data may be manipulated concurrently and the outcome of the execution depends upon the order of access? (1 point)

Part II. – Questions

1) What are the typical attributes in a PCB? (3 points)
(Describe them with one sentence for full mark!)

2) What are the preemptive and non-preemptive strategies for process scheduling? Shortly describe them! (4 points)

3) Describe the term "link" in the file system context! (2 points)

4) What is the absolute and the relative path? (2 points)

5) What is a system call? (1 point)

6) What is internal fragmentation? (2 points)

7) What is the readers-writers problem? (2 points)

8) What is the CPU–I/O Burst Cycle? (2 points)

9) Just as a file must be opened before it is used, a file system must be __________ before it can be available to processes on the system (1 point)

10) What are the three requirements a solution to the critical-section problem must satisfy? (Describe them with one or two sentences for full mark!) (6 points)

11) What is the term for describing the situation where a waiting process is never again able to change state, because the resources it has requested are held by other waiting processes?

______________________________ (1 point)

Evaluation

Part I. - Test : <u>at least seven (7) good answers</u> required for grading the exam!

Part II. - Questions:

0 – 12	failed
13 – 16	passed
17 – 19	average
20 – 22	good
23 – 26	excellent

Operating Systems – 2023 Fall

06. December 2023

Name: Onyekwere Chinedum-K. Neptun ID: H2EP85

Declaration

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Part I. – Test

1. A process can be terminated due to ________
   A) normal exit
   B) fatal error
   C) killed by another process
   D) all of the mentioned

2. If a deadlock is happened, the OS should always kill the process in the deadlock.
   A) true
   B) false

3. Which of the following scheduling algorithms use Time Quantum?
   C) FCFS
   D) SJF
   E) Round Robin
   F) Priority Scheduling

4. In operating system, each process has its own ________
   A) address space and global variables
   B) open files
   C) pending alarms, signals and signal handlers
   D) all of the mentioned

... can NOT be graded and counted in to my final grade.

Student's signature

Part I. – Test

1. Which of these might not be a job of the operating system? ________
   A) Stopping unauthorized users accessing files that do not belong to them.
   B) Handling HTTP requests.
   C) Allocating file handlers to programs that are running on the computer.
   D) all of the mentioned

2. In the Zero capacity queue ________.
   A) the queue can store at least one message
   B) the sender blocks until the receiver receives the message
   C) the sender keeps sending and the messages don't wait in the queue
   D) none of the mentioned

3. What is Turnaround time?
   A) the total waiting time for a process to finish execution
   B) the total time spent in the ready queue
   C) the total time spent in the running queue
   D) the total time from the completion till the submission of a process

4. In operating system, each process has its own ________
   A) address space and global variables
   B) open files
   C) pending alarms, signals and signal handlers
   D) all of the mentioned